

**What is the economic impact of transitioning to renewable energy
in emerging economies?**

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Abstract

The switch from fossil fuels to renewable energy has become the core issue on the global scale along with the economic repercussions that for the developing countries still remain quite complicated and so do the global developmental strategies. The impact of the energy transition on economic growth, energy security, employment shifts, industrial competitiveness, and fiscal stability in developing markets will be qualitatively assessed by means of literature-based analysis. From the paper, it is evident that there are opportunities like reduced dependence on energy imports, creation of jobs in the renewable sector, and the facilitating of innovation in technology as well as challenges that include the need for invested capital up front, the limitations of energy and the financial markets, and the possibility of uneven effects across regions or sectors during the transition. By pulling together the different lines of research that exist, the research points unequivocally to the fact that institutional capacity, policy coherence, financing mechanisms and global market dynamics are the major determinants of the overall economic outcome of renewable energy adoption. In conclusion, the paper asserts that the transition will bring about substantial long-term economic benefits; however, their realization in the context of emerging economies requires targeted policy design, sustained investment, and coordinated international support.

Introduction

The transition to clean energy sources is one of the siesta major and most widespread challenges that the 21st century will face. The global societies are not anymore reluctant to adopt the eco-friendly energy systems, therefore, they will have the climate change impact less intense, air pollution less, and to be more sustainable than before. The renewable energy technologies, namely wind, sun, water and battery storage, are those that are least harmful to

the environment and have reached a stage of economic viability as well. According to the most recent figures, a substantial portion of the newly installed renewable capacity is coming in at a lower price than the fossil fuel, thus, the clean energy transition is not only possible but also financially attractive (IRENA, as cited in Reuters, 2025). The transition from fossil fuels to renewable energy is not an only reshuffling of the energy markets but also a major influence on the investment patterns, politics, and economic development in such a way where both risks and opportunities are deeply opened up.

Nevertheless, it is the emerging economies that are the ones at the center of this global renewable energy drive that is more or less common everywhere. The transformation of the world's power supply to renewables, however, will be painful for these countries because they, together with the rest of the world, will be facing a unique mix of problems and at the same time will be under a certain degree of pressure. First, most of these economies are still in the process of rapid industrialization and their growth in terms of GDP is quite high. Their energy consumption is also growing very fast and this is mainly due to population increases, migration to urban areas, and higher individual consumption. The energy systems of these economies, which are less mature than the ones of the developed world, are not only required to switch over to cleaner sources but also to cater for the new demand in addition to that. Second, energy poverty is the main social and economic problem in many of these countries.

Billions of individuals in economies outside the OECD, according to Brookings, are deprived of modern energy services and therefore suffer in low- and middle-income countries where energy is not at hand at a price people can afford (Brookings, 2023). Financing constraints are a major barrier going, this way, a long way; renewable energy sources have made a turn-around and their costs are now lower but still, the high-interest rates, underdeveloped financial systems, and problems associated with less mature institutions found in the emerging markets are that they usually struggle with these conditions.

International Monetary Fund (IMF) has stated that the private sector's involvement in developing countries becomes tremendously essential if they are to clean up the energy sector and in particular, the IMF has indicated that these countries may need to invest more than the current trillions of dollars to develop their energy infrastructures.

The scenario presents an urgent and complex need for an analysis of the economic impacts caused by renewable energy shifts in developing countries. The financial/institutional problems of these nations not only add to the risks but also create a situation where their fast-developing industries will be finances not only for climate change but for huge overall economic development as well. What is more, due to the great concern energy poverty and equity issues raise, the results of the renewable energy transition will be closely linked with social development objectives as they will be fulfilling these larger goals by proving their effectiveness.

Thus, the present paper sets up the core research question: what would be the economic impact of the renewable energy shift in the developing countries? The question points to the area of “economic impact” and catches at both macro and micro angles, that is—investments, job changes, energy security, government budgets, and others — as factors in and resulting from the clean energy transition. The paper's design is to follow an interpretive, literature-based method. The authors depend on the interpretation of figures and the data amassed from the reviews of empirical studies and policy reports for renewable energy in developing countries. This interpretive methodology makes it easier to recognize the themes, the compromises, and the inconsistencies that may be covered underneath numerical forecasts.

The paper is structured in the following manner. First, in the Conceptual Background section, key concepts are explained—what is really meant by an energy transition, what is

defined as an “emerging economy,” and how one should approach economic impacts in this particular context. Next, the Literature Review goes through the previous findings of related research and discusses them under the headings of macroeconomic impacts, the atmosphere of employment, health of energy security, financial challenges, technological capacity, social equity, and policy frameworks. As an option for the reader, the paper then gives Comparative Perspectives by comparing the emerging economies with the developed countries, in order to pinpoint what is the specific character of the former. In the Discussion, I will combine the insights from the literature to support the claim as to which impacts will prevail, which barriers will be the toughest to overcome, and what the trade-offs will be.

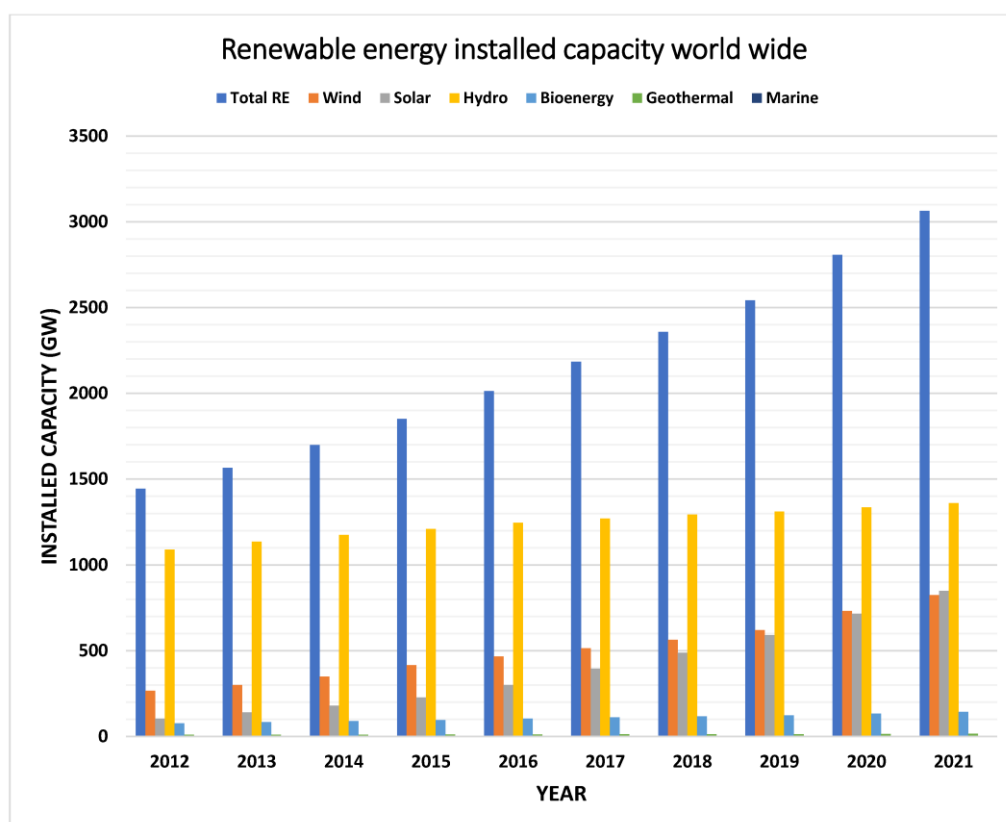
To begin with, the Conclusion will address the research question in a straightforward manner, expound the implications for policy, recognize the uncertainties involved, and recommend possible paths for new research. The overall picture is that the switch to renewables is a necessity for the whole world, yet the viewpoint of the developing countries gives it a different aspect that is composed of a variety of economic dynamics: high stakes, complicated constraints, and empowering possibilities. Through the interpretive framework of literature-based analysis, this paper would like to highlight a better comprehension of the shifts in sustainable development, the reasons behind it, the conditions that may lead to either the speeding up or faltering of this transition, and the future of sustainable development.

Conceptual Background

An unambiguous comprehension of the economic impact of renewable energy transitions in emerging economies necessitates concepts that are paramount to the understanding of the issue. These are the energy transitions themselves, the features of emerging economies, and the mechanisms of direct and indirect economic effects, which are

mostly associated with major shifts in energy systems. This part of the paper will prepare the conceptual frame for the analysis that will come next, it will root the discussion in well-established literature and at the same time, it will put renewable energy transitions in the context of wider economic transformations. By the term renewable energy transition, we talk about the steady movement away from the consumption of fossil fuels towards the establishment of a mixed energy system that is mainly composed of low-carbon and sustainable sources. From an economic point of view, the transition involves more than just changing technologies; it brings about a reordering of capital investment, infrastructure systems, production methods, and marketing surroundings.

Global renewable energy installed capacity by technology (2012–2021)



Source: Renewable Energy World/MDPI (2022). “Renewable Energy Installed Capacity Worldwide.”

Sovacool (2016), through his assertions, describes that the transition to sustainable energy is a multidimensional process that consists of a series of technological innovations, adjustments in regulations, and changes in the economic landscape. The transition starts with 'the n' m investments in the upcoming renewable technologies which are usually led by either policy incentives or cost competitiveness, and eventually, the whole process is carried out through the provision of the national grid. Electrical systems have to cope with the intermittency of renewable sources in the stages of the integration by improving the grid, providing storage, and managing the demand side (IRENA, 2021). The final stage, which is the scaling up to a large extent, sees a gradual substitution of renewables for fossil-fuel-based systems in terms of new capacity additions and thus influencing labor markets, industrial strategies, and even national budgets. Each stage has different economic impacts: the initial investment is important for the formation of capital, integration affects the prices and reliability of energy, and long-term scaling causes the productivity and the competitiveness of the industry to change.

Renewable energy sources are not a single type of technology; they include solar and wind but also hydroelectricity, geothermal power, and even the new proposal of green hydrogen, among others. To a great extent, these renewable technologies differ in their respective costs and that too, by a large margin, depending on the location, type of hybrid, etc. However, they do possess certain economic features which are quite similar and that makes them different from the fossil fuels-based system. One such feature is that the renewables have a tendency of having a more considerable upfront capital cost but a less expensive operation cost which changes the time profile of investment and returns (IEA, 2021). The increasing generation capacity from renewable energies and their adoption in the developing countries' grid where the potential for increasing the use of renewables is low will be made possible by their modularity. Some of these features will be not only the concern of

the energy planning of the respective countries but also the accompanying issues of job creation, fiscal policy, and long-term economic stability.

Emerging economies can be viewed as a heterogeneous cluster of nations at different stages of development and with different economic and political situations. Although the definitions differ from one another, the emerging economies are distinguished by slow but steady growth of the economy, development of the manufacturing sector, and active participation of the economy in the global market. In addition, they face problems such as limited access to financing and lack of proper institutional setup. The World Bank (2022) classifies these economies as middle-income countries that are undergoing a process of structural transformation but are still confronting major development issues. The banking systems of these countries are often not very deep when compared to the developed ones; hence, the borrowing costs are higher, and there is also greater vulnerability to macroeconomic shocks. All these factors together determine the viability and the cost of investment in large-scale renewable energy which in turn makes capital allocation a major issue. On top of that, emerging economies are often dealing with severe energy poverty, the International Energy Agency has projected that there are still hundreds of millions of people in Africa and Asia who do not have access to reliable electricity, therefore making it necessary to both increase the energy supply and at the same time decarbonize (IEA, 2023).

The energy demand pattern in the emerging markets economy adds to the necessity of the transition coming on a fast track. The global population concentration in cities and the growth of the industrial sector are the major drivers of energy demand and they are expected to contribute very significant increases in energy consumption in the coming decades. Accordingly, the countries with advanced economies would be able to replace existing fossil-fuel infrastructure by focusing their investments only on that sector, while the countries classified as emerging must invest in both infrastructures: the one that will be able to supply

the anticipated increased demand, plus the one that is environmentally friendly. It means that the economic impacts of the renewable energy transition are closely related to the development outcome in the broader context.

The successful integration of renewables can draw the line to energy independence, being less affected by fossil fuels' price swings, and promoting the long-term shift of industries. On the other hand, insufficient funding or wrong policy-making can result in the situation where the poor population gets poorer while the rich people become richer or even compel the governments to continue with the carbon-intensive development model.

The economic impacts of the transition from fossil fuels to renewable energy can be classified into three broad categories: direct impacts, indirect impacts, and externalities. The most immediate economic benefits are derived from the transition, such as the investment of capital in renewable energy and the employment of workers for the construction, installation, operation, and maintenance of the plants. It has been reported that the sectors using renewable resources for the generation of electricity are more labor-intensive per megawatt than their fossil-fuel counterparts, especially in the installation phase (Wei et al., 2010). During the transition of the emerging economies to renewable energy, the job effect can be considerable and it can lead to skill development and local manufacturing opportunities, however, the outcome varies a lot and depends on the structure of the labor market and the capacity of the supply chain.

Indirect economic impacts are those that have the longest time horizon but are also the most difficult to measure. The main areas of influence are the energy price that ultimately consumers pay, productivity, and general competitiveness of the economy. Improving the national trade balance due to the decrease in the imports of fossil fuels and giving the economy more resistance to external shocks, are among the advantages of the renewable

energy movement. The latter can also be the source of newer industries such as battery storage, electric mobility, and green hydrogen, and the positive effects of these industries can be seen in other areas as well. The International Renewable Energy Agency's (IRENA) analysis indicates that countries, which are heavily investing in renewables, gain the capability of innovation and thus, getting the economic productivity through efficient technology usage. These mentioned transformations have been noticed in the past, however, in the case of emerging economies, their impact is mainly determined by the domestic policy framework, the technological level of the industry, and the extent of participation in the global value chains.

Externalities consist of the environmental and public health benefits as the major economic impact category, namely: reduction in air pollution, lowering of greenhouse-gas emissions and cutting down of health care costs for treating respiratory diseases. The World Health Organization (WHO) has pointed out many times that air pollution is a major culprit of lost productivity and increased medical costs, which together account for a large economic burden on the developing countries. Fossil fuel dependence can be much reduced if a country opts for the renewable energy route which, in turn, will help in creating a more sustainable long-term economic growth path by cutting down on the mentioned economic costs. Even though difficult to measure these externalities represent a huge economic advantage of renewable energy adoption.

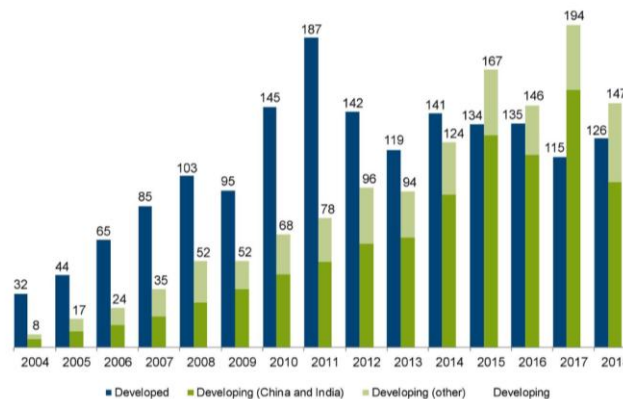
The above-mentioned concept elements combined show how complex the renewable energy transition has been in developing economies. They were not just the technological shifts coming along the way; these were comprehensive economic transformations that interfaced with the country's development priorities, financial structures, and institutional capacities. Thus, being aware of these concepts is a prerequisite for the understanding of the literature's evaluation of economic impacts and the continuing debates in the field.

Literature Review

Macroeconomic Effects

The macroeconomic consequences of the transition to renewable energy in emerging markets have been the subject of a lot of studies though the results differ according to the national contexts, the policy frameworks, and the technology trajectories. One of the most evident topics in the literature is that renewable energy investments can push the economy forward through more capital formation and the creation of new industrial activities. The empirical data provided by IRENA shows it very clearly that the adoption of renewable energy positively affects GDP growth mainly through investments being initiated not only in the energy sector but also in the manufacturing, construction and service sectors which have been IRENA's (2020) focus. The phenomenon of technology investments being distributed into different sectors is one of the reasons for the said economic effects, since the renewable energy sector has strong interconnections along the supply chain; solar and wind installations not only need a big share of local construction but also transportation and maintenance of the local service industries. In the case of developing countries, the interconnections are even more important since the potentials of renewable energy sources to attract industries into the locations with poor infrastructure are to a certain extent fueled by the renewable energy projects themselves.

Investment in renewable energy capacity in developed and developing countries (2004–2018)



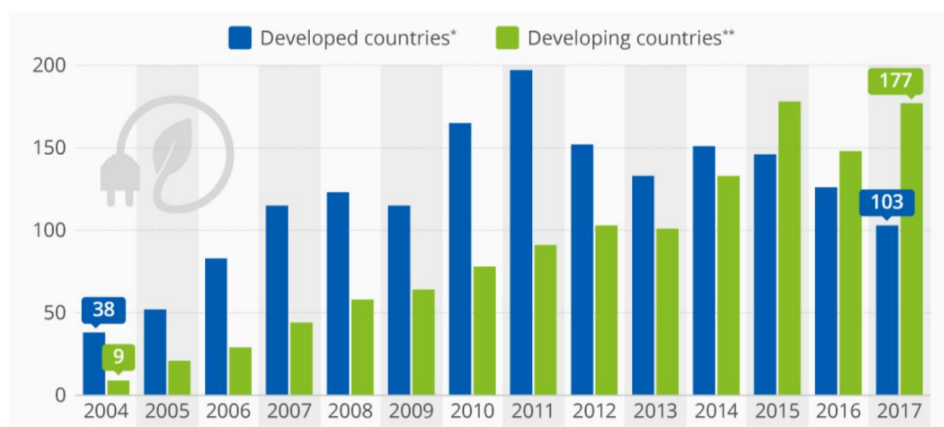
Source: UN Environment; Frankfurt School–UNEP Centre; BloombergNEF. “Global Trends in Renewable Energy Investment 2019.”

The spillovers of investment also come through the rising importance of renewables in the shaping of local capital markets. The cases of India, South Africa, and Brazil have evidenced that auctions and feed-in tariffs for renewable energy have contributed to the widening of green financing instruments such as green bonds and blended finance (Bhattacharya et al., 2020). These trends have the potential to finally lead to a reduction in the cost of capital through fostering investor confidence and hence, lowering perceived risks. However, the extent to which these microeconomic spillovers contribute to the strengthening of macroeconomic stability varies from one scenario to the other. To be more specific, the International Energy Agency (2021) indicates that the disparity in financing costs which is generally the practice with emerging economies may in fact act as a barrier to the full potential GDP mobility that is related to the renewable energy deployment scenario.

In addition, fiscal impacts are another major macroeconomic aspect. The transition to renewable energy taxes and government revenues in several ways, such as taxes on corporate

income from the new renewable industries, import duties on technology components, and the eventual reduction of fossil-fuel subsidies. A large number of studies support the view that the reform of subsidies alone can bring about a fiscal improvement in the balance of payments of emerging markets. The International Monetary Fund (2022) foresees that the complete abolition of fossil-fuel subsidies will lead to a considerable amount of public resources being released in places such as Indonesia, Nigeria, and Egypt, thus enabling the government to reallocate resources to areas such as social or developmental priorities. However, these fiscal benefits need to be weighed against short-term adjustment costs, such as payments to vulnerable households or retraining workers who have been laid off in the fossil fuel sector.

Comparative renewable energy investment by developed and developing economies (2004–2017)



Source: UNEP; Bloomberg New Energy Finance; Statista (2020). “Developing Countries Invest More in Renewable Energy.”

The trade balances are one of the factors that also get impacted by the renewables transition. A lot of emerging economies have a large-scale dependence on imported oil,

natural gas, or coal, which leads to their susceptibility to the price fluctuations in the global commodity markets. By producing a certain percentage of their energy needs from renewables instead of importing, the countries can not only improve their external accounts but also maintain stable foreign exchange positions. World Bank (2020) reports states that net energy importers such as India, Morocco, and Vietnam will have substantial trade balance improvements as a result of increased renewable penetration.

Nonetheless, the aforementioned benefits are still tied to the ability to identify parts of the renewable energy supply chain that are located in different places. In case the countries make a large use of solar panels or wind turbines that are imported from elsewhere, the benefits from trade can be neutralized by the corresponding increase in capital-goods imports which is usual during the initial stages of deployment.

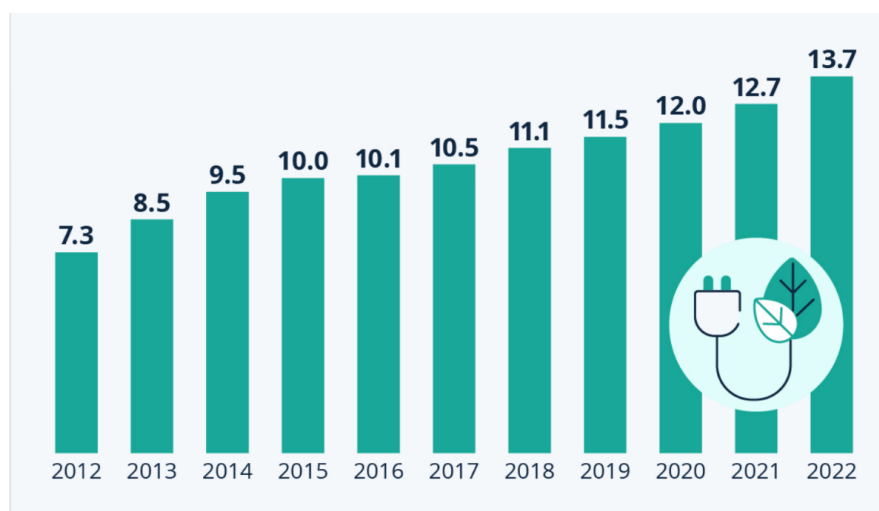
The general idea in the academic research is that the switch to renewable energy can give rise to considerable economic gains particularly in developing countries but the size of such gains will be determined to a great extent by the local production capability, the financial situation and the coherence of the policy measures. The studies have shown that there would be positive impacts on GDP, better fiscal stability, and more robust trading balances but the realization of these scenarios will not come cheap and thus require regular input from the institutions.

Employment and Industrial Development

The employment implications of energy transitions are complicated in emerging economies, as they comprise both the aspect of new job creation and that of job loss. Moreover, an often-quoted investigation conducted by Wei, Patadia, and Kammen (2010) claims that the renewable energy technologies normally give rise to a greater number of jobs per unit of electricity in the course of installation and construction phases than fossil-fuel

technologies. This conclusion has been reinforced by the latest research that looked at the emerging economies only, for instance, IRENA's Renewable Energy and Jobs reports not only document but also forecast that countries like Brazil, India, and China will continue to be at the forefront of renewable energy employment globally mainly due to the contributions from solar PV and bioenergy sectors to national job creation (IRENA, 2022). Such jobs are available at all levels of skills and activities ranging from manufacturing and installation to project development and grid integration.

Global employment in the renewable energy sector (2012–2022)



Source: IRENA (2023). “Renewable Energy and Jobs – Annual Review.” Statista chart

reproduction.

But job creation is not the whole picture. The transition also results in the end of jobs for many in the fossil-fuel industries, especially coal mining and thermal power generation. Research on South Africa’s coal-dependent areas, for instance, reveals that the labor-intensive nature of coal mining makes it socially and politically disruptive to lose workers (Ting &

Byrne, 2020). As a result, the challenge for developing countries is not only the creation of new jobs in renewable sectors but also the management of the decline of carbon industries in a manner that harms society the least. This situation has brought about the concept of “just transition” policies which provide skills training, income support, and regional development initiatives as a package deal.

Industry development is one more major aspect in the literature. The use of renewable energy can entice manufacturers in the domestic market by means of their sales in solar modules, wind components, inverters, and balance-of-system parts. China and India are among the countries that have already taken this opportunity to create competitive manufacturers with the help of industrial policies and technology-transfer initiatives (Mathews & Tan, 2015). However, there are still many emerging economies that are unable to take full advantage of these industrial benefits because of their limited technology skills, supply-chain problems, or the competition coming from established manufacturers. Researchers warn that if policymakers do not introduce complementary industrial strategies—like local-content requirements, R&D investments, and export-promotion efforts—the employment benefits of renewable energy might be limited to short-term installation work instead of long-term manufacturing segments.

The literature thus offers mixed messages: from one side the renewable energy transitions are the substantial employment generators and also capable of supporting industrial diversification, but from the other side those benefits depend greatly on policy design, and the readiness of the workforce, and the architecture of global value chains. Emerging economies which carefully coordinate between renewable energy deployment and industrial policy gain the job benefits that are more lasting.

Protection of Energy Supply and Price Stability

Among the many reasons for the adoption of renewables in developing countries, one of the most important reasons is energy security. Most of the countries for their energy needs are still importing the fossil fuels that leave them vulnerable to disruptions in supply and fluctuations in prices. The existing literature shows that one of the measures for strengthening energy security is the increase of the share of domestic production of renewable energy. The International Energy Agency (2020) in its report mentions that the emerging Asian economies like India and Thailand are mainly affected by the unstable global oil and gas prices. The generation of these countries can increase from renewable sources and this energy can be supplied to the domestic market making them less dependent on the outside world in terms of energy and also less affected by the tensions between the oil-buying and selling countries.

The use of renewables also means that there is not to worry too much about prices going up and down dramatically. Solar and wind technologies, though expensive to install, have low and predictable operating costs which are a very helpful factor in the case the electricity price is influenced by global commodity shocks. Many empirical studies point out that countries with a higher renewable energy share have less volatility in their wholesale electricity prices because of renewables providing a stable marginal-cost baseline (Gelabert et al., 2011).

In the case of developing countries, there is a direct relation between stability and the good industrial planning as well as energy affordability for households, in particular, during the times of global fossil-fuel price hikes. A good example of this is when the energy crisis of 2022, caused by the Russia-Ukraine conflict, affected the electricity tariffs in the countries with high renewables, such as India, where a lesser portion of the global price hike was transferred to the domestic electricity sector (IEA, 2022).

Intermittency is one of the major risks mentioned in the literature too. The non-availability of grid connectivity, storage facilities, and backup generation methods can cause short periods of instability in renewable energy supply. Thus, the long-term advantages of energy security and price stability are further dependent on parallel investments in power transmission networks, system balancing, and regulations reform.

Financing Challenges

Financing challenge is the barrier that keeps appearing all the time in the discussions around the deployment of renewable energy in the emerging economies. Capital costs are typically much higher in those countries than in the advanced economies, which largely impacts the affordability and consequently the bankability of the renewable projects. According to International Energy Agency (2021), one of the reasons is that the investors consider the risks involved with politics, currency, and regulations higher in the emerging markets and therefore expect higher returns, which leads to significantly higher financing costs for solar and wind projects as compared to Europe or North America. The difference can be substantial; for instance, solar energy developers in India or Brazil may have to pay interest rates that are 2-3 times higher as compared to those in Germany or the U.S. (IRENA, 2016). High capital costs undermine the economic appeal of renewables, particularly because these technologies require large upfront investment even though their operating costs are relatively low.

Risk perceptions are of equal importance in influencing the flow of funds. Generally, investors mention the instability of policies, the limitation of grid capacity, the drop in currency value, and the poor financial condition of state-owned utilities as the main things that, together, lead to the raising of country-risk premiums (World Bank, 2020). The study indicates that uncertainty regarding the issue of contract enforcement, power-purchase

agreement, and long-term tariff structures can even turn investor interest away from renewable energy technologies though they are cost-competitive (ESMAP, 2020). A large number of developing countries sell a big part of their generated power to state-run utilities; and the latter's financial conditions are often similar to those of South Africa's Eskom or India's distribution companies. Such vulnerabilities result in investor reluctance and hence, limit the size of renewable energy deployment.

Institutional barriers add more trouble to the already existing financial challenges. Lack of proper regulation in the institutions, poor administrative skills, and lack of coordination among the different policy aspects severely limit the growth of renewable energy markets. In some regions, the bureaucratic process in granting approvals for the project or land acquisition may lead to an increase in the project risk. Reports from the Climate Policy Initiative (CPI, 2018) indicate that although countries have set very high renewable-energy targets, they still lack the long-term policy stability necessary to attract sustained private investment, which is why concessional finance, blended finance arrangements, and multilateral development banks are important for reducing financing costs and risk. However, the amount of funding required still widely exceeds the amount that is currently flowing.

Technological Capacity and Innovation

The extent to which emerging economies can gain from the solar energy transition is largely determined by their technological capabilities. The transfer of technology has always been a major factor in the acceleration of the switch from fossil fuels to clean energy, through which countries have been able to obtain and modify foreign technologies using licensing, joint ventures, and participation in global value chains. The United Nations Environment Programme (UNEP, 2018) points out that not only China but also India and Brazil have been

very successful in utilizing the technology-transfer mechanisms to increase their domestic manufacturing of solar photovoltaics, wind parts, and biofuels. These countries are already making considerable investments in their domestic R&D, workforce training, and industrial parks which help them to create world-class industries for renewable energy.

On the contrary, many other developing countries still struggle to manufacture locally due to numerous obstacles. The lack of access to the latest technology, combined with weak industrial infrastructures, and low levels of investment in the area of research and development, limit the localization of renewable energy supply chains. According to IRENA (2017), the sub-Saharan African countries and some regions of Southeast Asia are still largely reliant on imported solar panels, wind turbines, and balance-of-system components. The transition to renewables can result in less power generation, especially in the high-skill manufacturing sector, where most of the production is concentrated in export countries. The dependency on imports can also be a factor in the trade balances going down during the initial period of deployment, even though the reduction of fossil-fuel imports may still be one of the long-term benefits.

The global value chains play a significant part in determining the innovation outcomes. The renewable energy industries are the closest to each other, with China dominating the solar module production and Europe being the leader in the wind-turbine technology. The emerging economies among which the investment in local manufacturing may be more observable are the ones that will be most likely to take advantage of these global networks by making their labor competitive, improving logistics, and providing an unchanging policy. On the contrary, those countries that do not bring their policies in line may end up in a subordinate position where they just consume the technology with no production. The relevant studies are in agreement that the industrial policy, which implies local-content requirements and designated subsidies, is one of the main drivers of the

technology capacity development (Mathews & Tan, 2015). However, such measures need to be applied very carefully to avoid cases of protectionism that incur large expenses without the actual development of capabilities.

Social and Distributional Impacts

The transition to renewable energy in emerging economies comes with considerable social implications especially in terms of the access to energy, equality, and rural development. There are still many countries that suffer from energy poverty, and thus, millions of people do not have access to reliable or affordable electricity. One of the most effective ways to provide electricity in remote areas is through off-grid and mini-grid renewable solutions, particularly solar home systems and small-scale hydropower. According to a report by the International Renewable Energy Agency (2019), decentralized renewable systems can not only supply electricity more cheaply but also faster than grid extension in most rural areas, thereby enhancing educational, health, and income-generating activities.

Inequality continues to be a major issue that is heavily debated in the social-distributional literature. The transition to renewable energy sources can lead to the creation of new opportunities, but the distribution of these benefits is not always fair. The rich are usually the ones who can buy and install all these modern technologies like solar panels, energy-efficient appliances, and electric cars, while the poor cannot even afford the cheaper options with the government subsidies. The prime concern of health care policy, according to the Overseas Development Institute's (ODI, 2020) research paper, is the carefulness of subsidy programs that could otherwise foster socioeconomic disparities in the society if low-income groups are not given priority. Besides, employment in coal mining regions would not be of any help for those areas unless the coal mining families get support in the form of training or otherwise.

The matter of the rural sector's development is another significant issue that needs addressing. Power projects such as wind farms, bioenergy facilities, and decentralized solar power can be blessings for the local communities as they will bring job opportunities and better infrastructure, apart from farmers being supported with electrified irrigation or cold-storage systems. Nonetheless, the researchers highlight the possible negative consequences of these projects: huge solar or wind power stations might encroach on the agricultural area, land ownership conflicts might arise or the profits might go to outside investors leaving the local area without any benefit (Baker & Sovacool, 2017). The lessons learned from these findings are that there is a crucial need for proper planning that involves the local community and for the establishment of equitable distribution of benefits systems.

Policy and Regulatory Dimensions

The policy and regulatory frameworks act as the primary determinants of investment, deployment, and the economic impacts of renewable energy. Among others, Feed-in Tariffs (FITs), Renewable Portfolio Standards (RPS), and targeted subsidies continue to be the most frequently examined policy instruments. In some developed countries, such as China, Turkey, and Thailand, FITs, which are essentially guaranteed prices for renewable electricity, were mainly responsible for the growth of the market for renewables in its earliest days. A world bank study (2015) asserts that FITs do lower the risks for investors by providing them with a reliable source of income, nevertheless badly set tariffs may have adverse effects on the government budget or cause boom-bust cycles in the renewable energy market.

In the meantime, the RPS and competitive bidding have become even more recognizable, especially in the case of India, Brazil, and South Africa. The regulations require the electric companies to buy a certain percent of their electricity from renewable sources, thus the renewable companies are competing in their pricing and the market is getting

demand that is predictable. The proof is out there that the auction method has led to the significant reduction of the cost of renewable energy, mainly through the process of making competition and transparency (IRENA, 2019) more active.

The subsidies are also very influential, nonetheless. In the case of many developing countries, the governments are providing support to the fossil fuel sector thus preventing the inflow of capital into the renewable energy sector, whereas others have offered to help in the form of financial aid for the adoption of renewables through tax credits, soft loans, and capital grants. The IMF (2022) posits that reconsidering the subsidy structure is a prerequisite to be able to meet the dual objectives of fiscal health and energy transition.

Governance problems are not separated from policy designing. The private sector will be attracted if the rules and regulations are very transparent, the permitting systems are very efficient, and the enforcement mechanisms are strong. Poor governance, red tape, and inconsistency in policy can drive investors away and slow down the approval of projects. Research states that the countries with long and clear energy strategies and good institutions, like Chile and Morocco, are the ones that get the most renewable investments that are less volatile (ESCWA, 2019).

Comparative Perspectives

The comparison of different countries reveals that the economic effect of the switch to renewable energy varies greatly among the emerging economies and this is determined by the amount of resources available, the institutional capabilities, the industrial foundations and the policy frameworks. These differences show that the changing of the energy sector to renewable sources, although it has broad economic opportunities, does not bring about the same results everywhere; instead, the outcomes depend on the specific national pathways and

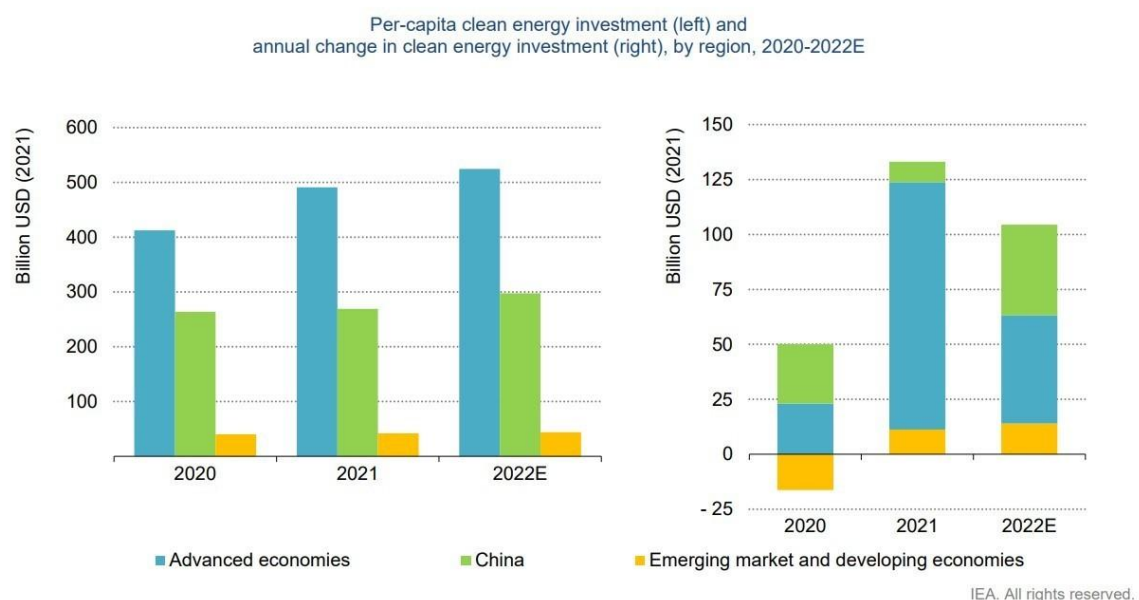
the respective choices of policies. A comparison of the countries like China, India, Brazil, South Africa, Vietnam, and Morocco, clearly shows that there are certain patterns that can be used to explain why some emerging economies have received a more considerable share of the economic profits from the renewables than others.

Among all the countries, China has shown the greatest ability to access the benefits of the renewable transition for the purposes of upgrading its industries and of positioning itself as a global leader in the market. Its gigantic state-investment, the enormous capabilities of its domestic manufacturing, and the clear policy support have made China, the world's largest supplier of solar photovoltaic (PV) and one of the leading exporters of wind components (IEA, 2023). The success of this nation has been made possible by coordination of industrial policies, large-scale public financing, and strong technological absorption capacity. Hence, the renewable transition has been a major factor in the growth of China's GDP, the creation of jobs, and the expansion of exports. Other emerging economies with weaker industrial infrastructures, on the other hand, have not been able to partake in the same manufacturing benefits even if they have increased their renewable energy deployment.

The Indian case presents another route which is influenced by the market-oriented factors and policy trials. The competitive bidding process has resulted in a significant reduction in renewable energy prices, thus making India one of the fastest-growing solar markets in the world (IRENA, 2020). However, the high interest rates, state-wise regulatory confusion, and the financial woes of the power distribution companies have all served to limit the economic benefits of renewables from being realized to the greatest extent. Furthermore, India's case also brings to the forefront the challenges posed by local manufacturing in the face of already established global value chains; despite government incentives, the country still relies heavily on imports for PV modules. On the other hand, India has been able to provide the economy with a large number of jobs in the areas of installation, maintenance,

and rooftop solar—all while cutting down on the import of fossil fuels. Brazil is an example of a resource-centered model that has the advantage of being able to make use of the wind as a resource, a large hydropower capacity, and an established biofuels market. Although Brazil has been able to make the transition to renewable energy relatively successfully, one of the factors is the abundance of the earth's natural resources and the variety of clean energy sources.

Clean energy investment by region (2020–2022)



Source: International Energy Agency (IEA). “World Energy Investment 2022.”

Research has indicated that Brazil's wind sector has positively impacted the economic development of the northeast region through the creation of manufacturing and logistics activity clusters (EPE, 2021). However, Brazil's economic volatility and changes in political priorities have been the main reasons for its intermittent investment in wind power, which, in turn, shows the dependency of renewable transitions on macroeconomic stability. Brazil's

experience underscores the importance of political continuity and policy stability for sustaining long-term economic benefits.

In Africa, the contrast is vivid between South Africa and Morocco regarding the pathways they have chosen. The renewable-energy procurement program of South Africa (REIPPPP) did initially draw in global investments of considerable amounts due to its fair and very competitive auction design, but the financial problems of Eskom and uncertainty of regulations have been the main factors responsible for slowing down the momentum (Baker, 2018). The other South African potential of wind and solar must stay untapped due to the limitations of the existing grid and socio-political factors, thus the whole economy cannot benefit from it. On the flip side, Morocco has often been praised for its efficient administration and proper planning that helped it to conquer the market and become the leader in solar energy in the region, hence, the Noor Ouarzazate complex (World Bank, 2019). The industrial strategy of Morocco has not only attracted the attention of the international community but also of the local players in the sense that components like cables and mounting structures are now being localized even though the high-tech parts are still imported.

The story of smaller emerging economies like Vietnam and Indonesia is one of the policies that need to be quick to adapt as well as the investment climate. A rapid solar boom in Vietnam occurred from 2018 to 2021 driven by the government's decision to introduce the very favorable feed-in tariff system which in turn, led to one of the fastest solar expansion worldwide (ADB, 2022). Korea, however, experienced grid congestion and sudden policy changes that turned the tide against it; hence it was a classic scenario where ill-timed policies hurt rather than help the economic growth. Indonesia has been limited in its coal usage, difficult regulatory situation, and lack of incentives for the clean energy sector. The country

has geothermal resources but mainly due to slow licensing process and high risk, the investments have been very limited.

In summary, using different countries' views, it is substantiated that renewable energy transitions in developing nations cause substantial economic changes mainly in those places where the government combines stable policies, good institutions, financing at competitive rates, and strategic industry policies. Through the adoption of renewables as a part of their economic development plans—China, India (partly), Brazil (regionally), and Morocco—they are the ones who benefit the most in the long run. On the other hand, countries that face frequent changes in regulations, have not so good grid infrastructures, high financing costs, or lack of industrial skills seldom find it easy to turn renewable energy deployment into steady economic benefits. Such comparisons prove that the transition to renewable energy is not only about technology but also about governance, industry, and development challenges requiring coherent state capacity and a long-term vision.

Discussion

The changeover to renewable energy in developing countries has economic effects that are not only complex but are also influenced by natural phenomena (structural factors), the existing institutional capacity and the overall political economy where the energy systems are operating. The macroeconomic effects, industrial development, energy security, financing constraints, and others that have been reviewed in the literature have shown a pattern of discrepancies that can still lead to the usually expected economic consequences. The findings above are synthesized and the major economic impacts in the developing world, the most confining bottlenecks, and the dilemma that governments need to tackle are presented as the area of cleaner energy transition in which they try to speed up the good. The differences in

political economy from governance strength to the power of vested interests are also highlighted as the factors that determine the economic benefits' distribution.

The literature clearly shows that in the case of emerging economies the renewable energy transition represented an economic impact that was first coming through in terms of costs and only later in terms of benefits. The deployment of renewables is usually associated with large front investments, especially for capital-intensive technologies such as solar PV, onshore and offshore wind, and grid upgrades (IRENA, 2021). Industrialization, job creation, and tax revenue are features that come gradually over time. This mismatch of timing is a source of political and financial tensions, particularly in countries with limited fiscal space where policymakers have to deal with competing priorities like poverty alleviation and infrastructure development. If we look at studies conducted in India, South Africa, and Indonesia, we find that governments continuous dialogue with the macroeconomic benefits projected over decades, often hesitating to make the immediate financial commitment that they would otherwise have to do (Bhattacharya & Sen, 2021; Baker, 2018). Thus, the renewable transition can be regarded as overlapping with broader development issues, thereby limiting the economic impact to those long-term planning horizons that are not always politically feasible.

Among the various economic impacts, financing constraints consistently appear as the most binding barrier. Emerging economies face systematically higher costs of capital compared to developed markets due to perceptions of regulatory risk, currency volatility, and limited depth in domestic financial markets. IRENA (2023) states that the weighted average cost of capital for renewable projects in Africa and some parts of South Asia could be at least twice and up to three times higher than that for Europe. This difference in financial terms makes it very hard for the regions to have any renewable sources of energy as the cost will be so high that only the initial project application would be affordable. For instance, developers

in Nigeria, Kenya, or Bangladesh still cope with high capital costs that cut down the profits of investment even though the solar PV technology has become cheaper in the world. This is why countries with bigger capital markets or strong state-backed financing - China and to a lesser extent India - have been able to deploy faster than others.

Institutional quality and policy consistency are tightly linked to the financing issues, which in some instances may hold back the deployment of renewables. The regulatory framework and the government's commitment to renewable energy are among the factors that the existing literature identifies as the foundations of long-term economic impacts of such transitions (Bhamidipati & Sagar, 2021). Initially, South Africa's REIPPPP program attracted massive investment because of its transparent auction model, but later political uncertainty and financial problems at the local utility turned the situation around making investors skeptical (Baker, 2018). On the other hand, the energy strategy of Morocco is referred to as a model of stable governance and coherent long-term planning, which has allowed the country to access international funding for huge solar parks that represent some of the largest in the world (World Bank, 2019).

These differences highlight that the institutional strength greatly influences the level to which renewable investments lead to broader economic gains even when the resource endowments are almost the same. One more subject that the literature points out to as a major theme is the industrial development and the local value capture extent. The reduction of renewable technology costs worldwide that has been happening thus far is good for developing countries as they are the consumers, but at the same time, it is putting the domestic manufacturing under the gun because imports from the firms that are on the top of the world value chain - mainly China, Taiwan and a fast-growing Southeast Asia - are so cheap that local producers cannot compete with them (IEA, 2023). This interaction defines the size of the economic spillover. For example, Brazil, Turkey and India have engaged in

building domestic manufacturing through import tariffs and local content requirements but with varying results. Some researchers have pointed out that the adoption of such regulations might lead to higher costs of technology and the establishment of industries that are completely protected but remain uncompetitive globally, whereas at the same time job creation and industrial cluster formation can be the other outcomes of such measures (Lema & Rabellotti, 2020). Therefore, local manufacturing has been successful in the mid- or downstream segments mostly, while the import of high-tech parts such as semiconductors or advanced inverters continues. The more extensive implication is that renewable energy can be a factor for economic diversification, but its potential for large-scale industrial upgrading is heavily reliant on the existing technological capabilities and the innovation ecosystems' strength.

Technological capability has a fundamental interaction with the reallocation of knowledge and innovation. Future up-and-coming economies are going to be arguing that they will be benefiting more from the adoption and diffusion of existing technologies rather than developing the frontier technologies independently (Altenburg & Rodrik, 2017). Exporting of hardware, taking part in partnerships, entering into joint ventures, and obtaining licenses serve as avenues for new local firms to be capacitated to absorb the coming innovations. One instance of this is the confluence of China's initial venture into the wind energy technology and its collaboration with the European manufacturers which gave the Chinese companies *par la carte* access to transferring plus scaling up of the tech knowledge before producing China's wind energy (Lewis, 2013). Analogous reasons have been given in the case of the Indian and South African renewables, though the results have been different. The current level of technology in the respective countries acts as a multiplier: to put it more simply, to the extent that the innovation system is weak, the renewable transition will yield

fewer indirect economic benefits. Thus, reinforcing the need for coordinated policies on industrial and educational fronts.

Another important aspect of the literature is the social and distributional dimension of economic impacts. Socially responsible corporations that invest in areas of renewable energy will not only enhance their image publicly but also get a better ear to the needs of their less fortunate customers through local presence. Access to energy, lower fuel costs for households, and creation of employment in rural or less developed areas are a few direct and indirect ways through which renewable energy transitions can be impactful on the social front. The example of off-grid solar and mini-grids has been so notable in terms of positive impact on rural economic activity in East African and South Asian countries (Hassan & Lucchino, 2016). The upsides of the projects, however, are not across the board. Displacement of local residents due to large renewable projects and employment getting disproportionately concentrated to already skilled workers are some of the scenarios under which large-scale renewable projects may cause inequality.

Additionally, changes in jobs from fossil fuels to renewables are hardly ever smooth. Research on coal-reliant areas in India and South Africa shows that without proper retraining programs, a large number of workers will lose their jobs, which is a major risk (Yanguas-Parra et al., 2019). Therefore, the overall economic result is contingent on the government's choice of implementing policies that are inclusive and prepare for labor-market disruptions and consequently assist the affected populations.

Moreover, energy security and price stability are definitely the main issues highlighted. Numerous developing economies face the global oil price fluctuations otherwise they will suffer losses. The war between Russia and Ukraine that resulted in energy crises put their economic case for internally produced renewable energy sources as a buffer against

outside disruptions stronger (IEA, 2023). Countries like India, Vietnam, and the Philippines have recently altered their policies referencing this rationale. However, it is important to note that energy security advantages do not come automatically; they are dependent on the additional investments made in grid infrastructure, storage, and system integration. Otherwise, the high percentage of intermittent renewables may lead to a new kind of instability, just like what happened in Vietnam with grid congestion due to its rapid solar expansion (ADB, 2022). In this regard, it is correct to say that energy security benefits only come about when deployment is in line with wider system planning. The discussion of these topics leads to the conclusion that the renewable transition in developing countries involves a number of economic trade-offs.

Governments must balance the short-term financial demands with the long-term economic and environmental benefits, they must choose between importation at low prices and the establishment of local industries; and they must also reconcile the requirement of quick deployment with the need for ensuring fairness among communities. All these trade-offs set the contours of the political viability of renewable energy policies. For example, if the use of subsidies or feed-in tariffs leads to increased public expenditure, then the pressure on the government may lead to public outcry, despite strong arguments for long-term benefits. On the other hand, if a government forces strict local-content requirements in the industrial sector to uplift the industry, the consumers may be met with expensive energy, which in turn results in delayed deployment and non-achievement of climate goals.

The political economy factors among others determine the outcomes, with the dominance of the fossil fuel industries being one of the factors. In a considerable number of developing countries, the energy sector is dominated by state-owned companies, and the interests of these companies may result in delays of or distortions in the use of renewables. The case studies of South Africa's Eskom, the coal industry in India, and the utilities in

Indonesia show that the actors that have been in the market for long periods usually resist reforms, which they see as a threat to their market share, (Baker, 2018; Bhattacharya & Sen, 2021). This leads to the emergence of incoherent policies and the slow movement of investments. Meanwhile, countries like China that have made it a point to link the use of renewable energy to the national growth stories in a dedicated way, frequently, show more uniform backing of policies and better-defined industrial strategies. The ruling context on a larger scale thus becomes a crucial determinant in the allocation and amount of the economic impacts.

The literature when viewed as a whole suggests that the transition to renewable energy is not only an invitation but also an opening for the emerging economies to utilize their investments to the fullest economically, however, the outcome will depend on the factors of institutional capacity, financing environments, industrial policies, and socio-political dynamics. There is no doubt that the transition has economic gains as a consequence and they are not uniformly distributed; on the contrary, they are most marked in the nations where the governments have already linked energy and development goals through consistent and long-term strategies. Such strategies may include among others: the establishment of regulatory frameworks that are reliable, the lowering of financing-related risks, the construction and upgrading of the electricity grid, the transfer of technology, and the assurance that the transitions are inclusive of all parts of society. Countries that artfully manage these factors not only gain access to cleaner energy systems but also enjoy enhanced economic diversification, energy security, and resilience to external shocks among other advantages.

Conclusion

The transition to renewable energy by emerging economies is a significant economic shift that extends beyond the energy sector. The research question addressed by this paper was: What is the economic impact of transitioning to renewable energy in emerging economies? Based on extensive literature review and analysis of the open-access literature, the study concludes that together with the immediate benefits, the adoption of renewable energy brings various economic benefits like long-term growth, diversification of industries, energy supply security, and social development. However, these benefits are, to a large extent, uncertain and not evenly distributed. They rather depend on the local conditions, institutional capacities, and the overall policy environment in which the transition takes place.

Almost all literature agreed on one point, that is, the transition to renewables in developing countries is always correlated with significant initial costs. Besides the upgrades of transmission networks, storage systems, and grid management infrastructure among others, the cost of the renewable technologies is also added (IRENA, 2021). The economic benefits in the long run—like job creation, reduced fossil-fuel imports, cleaner industrial development, and lower electricity costs—are major but often come in small doses. This time lag forces governments to endure the fiscal burden of the moment while betting on the late but lasting gains. Morocco and China, thanks to their strong institutions and coherent policy frameworks, have been the countries that managed to cope with the trade-off and reap the economic benefits in the long run (World Bank, 2019; IEA, 2023).

Furthermore, the findings highlight that over half of the emerging markets still experience the tightest barrier financing as a main challenge. The costly borrowing, the small amount of low-interest loans, and the high-risk perception are still the main hindrances of getting renewables at a faster pace. In its report, IRENA (2023) mentions that the cost of

renewable energy projects in most African and South Asian economies is several times higher than that in developed countries due to the poor financing conditions. Thus, the international climate finance, blended finance mechanisms, and domestic financial reforms will all help in the elimination of the financing gap in which the broader economic opportunities can be realized.

One more main point is draught up by the analysis regarding the disparity in capabilities among the emerging markets to reap the industrial and technological gift of the new investing area. Renewable energy can trigger new manufacturing sectors and skillful employment, but the measure of these profits is very much related to the existing technological know-how and the inviting industrial policies. Countries that do not have strong innovation ecosystems or that keep changing their policy signals often find it difficult to create strong domestic industries and consequently, depend on importation of technology (Lema & Rabellotti, 2020). Thus, the whole economic impact of the renewable transition shifts depending on how governments view the sector in their overall development planning.

The last thing to mention here is that the discussion brings attention to social and distributional considerations as a very important factor. Renewable energy transitions promise to bring the whole population of a country or a region into the modern energy net, improve the living conditions of the people in rural areas, and open up new labor markets in the places that were previously considered unserviceable for workforce purposes but still the literature shows that these are not the issues, which will just happen automatically. In the absence of affirmative measures that would spoil labor-market disruptions, boost community engagement, and secure everybody's access to new job opportunities, transitions may only contribute to the already existing inequalities (Hassan & Lucchino, 2016). Accordingly, socially inclusive policy design becomes a necessity for the generation of economic welfare.

To conclude, the economic transformation through the use of renewable energy in developing countries is huge but depends on some conditions. In cases when the institutional capacity is strong, financing is easy to access, and the policies are coherent, renewable energy can act as a catalyst for economic transformation and long-term resilience. On the other hand, if these enabling factors are weak, the effects are mostly dispersed or not at all. Therefore, future research should consider comparative studies and longitudinal studies as they would give a better understanding of the political, financial, and technological dynamics that affect the different economic outcomes of energy transitions in various emerging-market contexts.

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